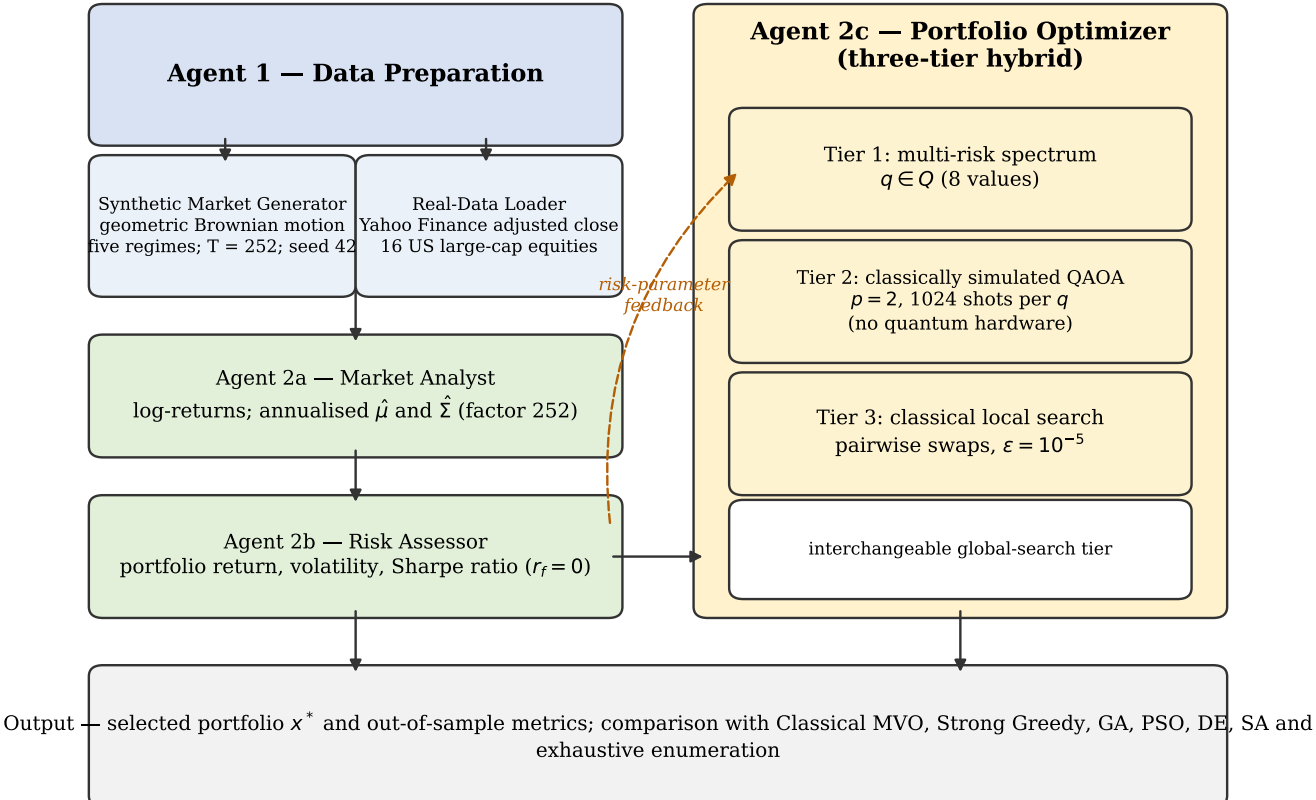
Agent 1 — Data Preparation
Synthetic Market Generator
geometric Brownian motion
five regimes; T = 252; seed 42
Real-Data Loader
Yahoo Finance adjusted close
16 US large-cap equities
Agent 2a — Market Analyst
log-returns; annualised $\hat{\mu}$ and $\hat{\Sigma}$ (factor 252)
Agent 2b — Risk Assessor
portfolio return, volatility, Sharpe ratio ($r_f = 0$)
risk-parameter
feedback
Agent 2c — Portfolio Optimizer
(three-tier hybrid)
Tier 1: multi-risk spectrum
$q \in Q$ (8 values)
Tier 2: classically simulated QAOA
$p = 2$, 1024 shots per q
(no quantum hardware)
Tier 3: classical local search
pairwise swaps, $\varepsilon = 10^{-5}$
interchangeable global-search tier
Output — selected portfolio x^* and out-of-sample metrics; comparison with Classical MVO, Strong Greedy, GA, PSO, DE, SA and exhaustive enumeration